

SKINNER'S SCIENTIFIC SPECTRUM

“The question to ask is not now ‘What is everything made up of?’ but rather the more subtle ‘How is everything made up?’

- NEO SKINNER

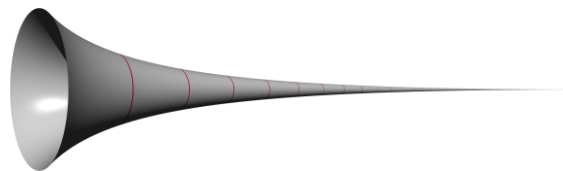
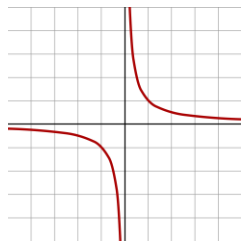
MATHEMATICS

THE PAINTER REALLY SHOULD RECONSIDER...

Last month we discussed the fascinating world of non-Euclidean geometries, and luckily for you, we are not quite done yet! We focused on three main planes: flat, spherical, and hyperbolic - and it is the final that is the most interesting. Gabriel's horn, the specific example of hyperbolic geometry we studied, is also intriguing as a shape.

Consider this, what defines a three-dimensional shape? Most people would probably agree that it consists of, here goes, an enclosed space of measurable volume with sides formed from a series of lines meeting at vertices and of definite surface area. If this or similar is your answer - you are not alone! This is how I too would define a 3D shape, well, I *would* if it were not for Gabriel's horn.

You see, Gabriel's horn exhibits some truly extraordinary properties which defy belief. As the volume formed when a reciprocal graph (that is, $\frac{1}{x}$) is rotated about the x -axis, it naturally follows that the shape has an infinite surface area due to its infinite extension in the x direction - as the shape is only defined as starting at $x = 1$ (ask Torricelli, *not* me).



One Revolution; LTR: Reciprocal Graph, Gabriel's Horn

What is more difficult to see - and where the story really begins - is that the volume is actually finite!

Let us start by defining a couple of key items; first, the volume of revolution - the technical term for what we are finding here - is π times the integral of y squared. We will begin with the integral first and then do the multiplication. y squared is just $\frac{1}{x^2}$ and this integrated gives us negative $\frac{1}{x}$, neat!

$$\pi \int_b^a y^2 dx$$

Finding the Volume of Revolution



Next, we must evaluate our integral with the bounds given to us: 1 and positive infinity. For the infinite case, we must use a limit, and it probably will not take me much persuasion that if we put 1 over infinity we are finding a very small quantity indeed – in fact, the limit gives us 0: $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Then, for 1, negative 1 over 1 is just negative 1 ($\frac{-1}{1} = -1$). To evaluate the integral with the bounds, we take the lower bound from the higher bound and get $0 - -1 = 1$.

$$\pi \int_1^{\infty} \frac{1}{x^2} dx = \pi$$

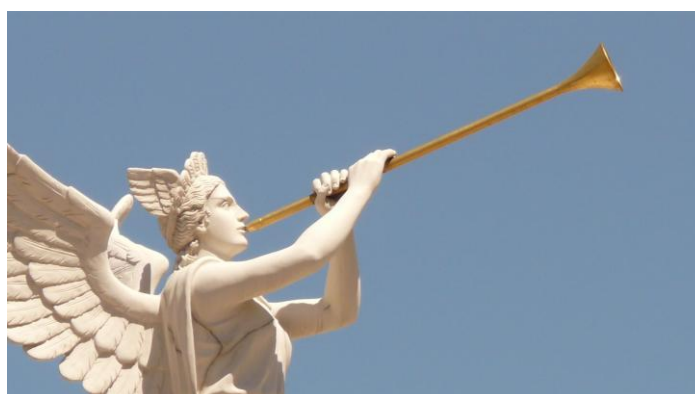
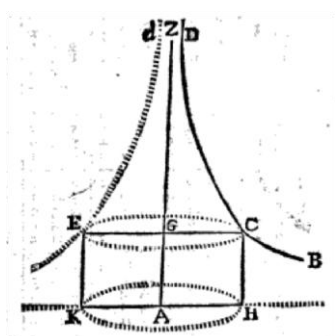
The Volume of Revolution of the Reciprocal Function

Multiplying out by π , the result is that the volume of Gabriel's horn is simply π units.

If you now want more proof for the infinite surface area result – and I do not blame you if you do – turn your eyes to this: the curve of the reciprocal function extends infinitely far as it has the x -axis as an asymptote that it will never touch. As it extends infinitely far, regardless of what we use to approximate the surface area – whether it be a cylinder, cone, or something else – the surface area will always be infinite.

So this is where the strange result comes from. I have tried here to explain this extraordinary paradox without using too much mathematics and I hope those of you who are confident can follow the maths in detail – and if you are less confident then I hope you followed the main reasoning if not the mathematical basis for the steps. Your definition of a 3D shape should by now have changed, the surface area does not necessarily need to be measurable...

In 1644, Evangelista Torricelli – who I alluded to earlier – stumbled across this result using a predecessor to modern calculus and it later developed into what is now called the painter's paradox; a finite volume of paint is required to fill Gabriel's horn, completely coating the inside surface, but yet it would appear an infinite quantity of paint is required to coat the outer surface!



Gabriel's Horn; LTR: Torricelli's Workings, Statue of the Archangel Gabriel (Complete With Horn)

This is actually not a paradox, but I will leave it as an exercise to you to prove *why*. Hint: consider how you can make the surface area change for a set volume, is it always the same?



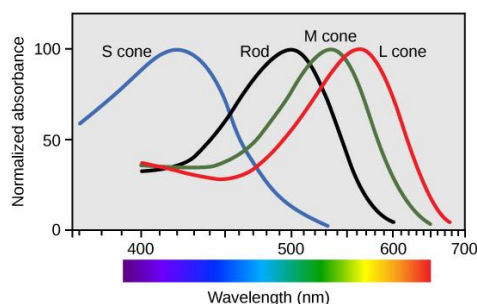
PHYSICS

NOT QUITE A RAINBOW

On the 6th of February 1672, a prism was turned towards a beam of sunlight and the most extraordinary thing was observed, a spectrum of seven colours emerged: first red, then orange, yellow and green, followed by blue, indigo and finally violet. Isaac Newton, holding the prism, would rise to fame and infamy from this very occasion.

We have all heard this story, and it can be found in the *Philosophical Transactions of the Royal Society* from February 19th 1672 (Newton, 1672). But, how does it happen?

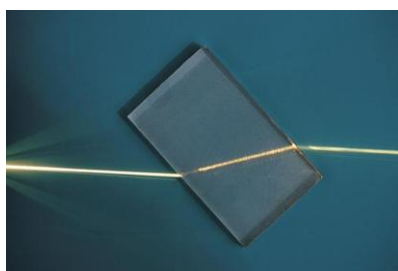
‘White’ light from the Sun is comprised of a continuous range of colours – the red to violet that we are familiar with – and it is the combination of these colours that produces the appearance of white in our eye as the colour-receptive cone cells of red, green and blue in our retinae are equally stimulated. As the three types of cell are equally stimulated, our brain interprets the colour as a mixture of all three, or white.



The Equal Stimulation of Retinal Cones to White Light

This explains the resulting spectrum, now for how it is produced. We can discuss colour in terms of wavelength, whereby a light wave with a specific wavelength has a specific colour – for example, red light has a wavelength of 700 nm (nanometres) – and, due to the wave speed equation ($v = f\lambda$), we can also assign a frequency to specific colours.

When light travels through different materials, it does so at different speeds. We describe this as the refractive index of the material, and it can be calculated as the ratio of the speed of light in a vacuum against that of the material ($n = \frac{c}{v}$). Using this, we can then calculate the angle of refraction using Snell's Law: $n_1 \sin \vartheta_1 = n_2 \sin \vartheta_2$, which can be rearranged to give us $\vartheta_2 = \sin^{-1} \frac{n_1 \sin \vartheta_1}{n_2}$.



Light Refraction in a Glass Block

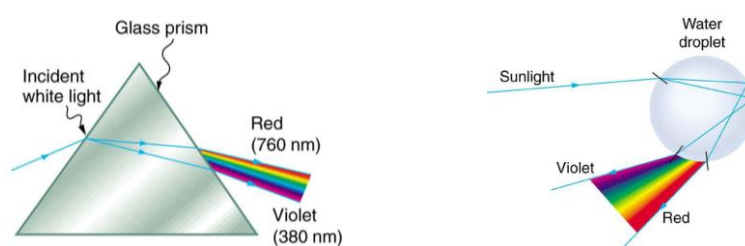


If you did not quite follow that, all you need to know is that light bends through an angle when it enters into different materials – a phenomenon known as refraction. Now, here comes the key link; the refractive index is unique not just to the material, but also to the wavelength undergoing refraction. This means that, for the continuous range of wavelengths that make up white light, there is also a continuous range of refractive indices.

Putting all of this together, as white light travels through one of Newton's prisms, red light has a lower refractive index (with longer wavelengths) and is refracted less than the shorter wavelength (and thus higher refractive index) violet light. This causes the spectrum of light to emerge from the dispersion of white light.

This I conceive, is enough for an Introduction to Experiments of this kind...

(Newton, 1672)



‘Look Mum, a Rainbow!’; LTR: Dispersion of Light in a Prism, Dispersion in a Water Droplet

Dispersion using a prism might be something we are all familiar with, but something that we have definitely all seen is a rainbow – and this uses dispersion too! Instead of a prism, sunlight refracts to enter a raindrop, bounces off the back and refracts again on exit; with the net effect being a beautiful rainbow. This fact also means that you can only ever see a rainbow when the Sun is behind you – as light has to bounce back into your eye.

The final question many people have about rainbows is: ‘What the heck is indigo?’ Well, short of being a heavy navy blue (The Editors of Encyclopedia Britannica, 2024), there is good reason to suspect that Newton actually introduced, not one, but two colours to the rainbow spectrum we now recognise to make it up to the number seven. There has long been a rumour that orange and indigo were introduced to allow the number of colours to be the same as the number of days in a week and the number of then-discovered planets. The number seven took on mystical significance and so Newton just bolted on these two extra colours to please the contemporary audience – not quite a rainbow!

Instead of ‘Richard Of York Gave Battle In Vain’ you could now try ‘Rainbows Yell Get Better Vision!’

But, that is only mere hearsay and there is little written evidence that Newton *actually* did add extra colours, although it makes for a nice story and taking it as ‘gospel’ does avoid the hassle of having to explain to five-year-olds what the colour ‘in dee go’ really is – there are *no* excuses for difficult explanations!

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CHEMISTRY

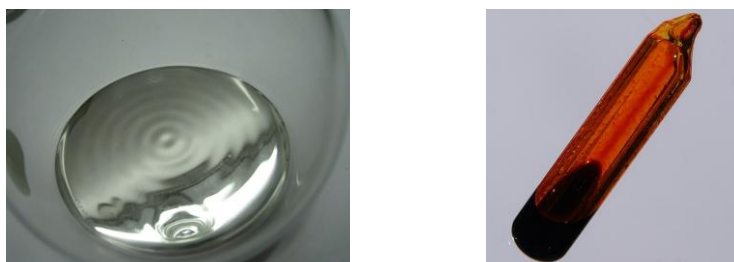
ESPRESSO METAL-INI

A relaxing fireside, just you and a cup of hot cocoa. Until your spoon vanishes in your hand! Turns out, a mischievous friend just switched your spoon for one made of the metal gallium – which melts at just 30 °C (Skinner, 2023b), what a prank!

But, we are not here for pranks today – we are here for metals that are *always* a liquid in nature. That is, a liquid at a standard temperature of 293.15 K (Kelvins, or 20 °C) and an atmosphere of 101 325 Pa (Pascals) (Skinner, 2023a/b) (defined as STP) – that got very specific, very quickly!

And for good reason, we have to define things very precisely as some elements may only change state at close to our defined standard conditions, meaning that someone in Morocco might say that something is a gas whilst in Sweden it remains a liquid!

So, the liquid elements are – drum roll please – bromine, mercury and (maybe) copernicium and flerovium, by our exacting standards. Mercury is the most stable of these, with bromine a volatile liquid-gas mixture in most cases, whilst copernicium and flerovium are not yet confirmed (more on that later).



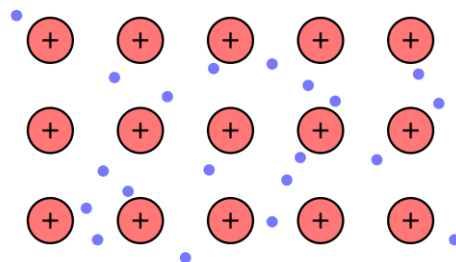
Refreshing; LTR: Liquid Mercury, Volatile Bromine Liquid and Vapour

Now we know the leading cast of this tale, it is time to dive into the science of *why* they are liquids at STP. We will focus on mercury first as it breaks the mould more than bromine; in solid mercury, there is metallic bonding, where metal atoms lose electrons which become delocalised and allow for a force of electrostatic (fancy word for ‘charge’) attraction between the metal atoms (now ions) and the unbound electrons – holding the structure together.

However, solid mercury only forms below -38 °C (Skinner, 2023b) so the metallic bond is weak at STP. This is because of the electronic structure of mercury – how its electrons are arranged to those non-chemists – which is $[\text{Kr}] 4d^{10} 4f^4 5s^2 5p^6 5d^{10} 6s^2$, the full $4f$ subshell provides less shielding for the outermost electrons and this can be seen in the ‘lanthanide contraction’ whereby the atomic radius shrinks as the atomic number increases. This contraction means that it is harder for the atoms to lose electrons.

In mercury, this effect is compounded by the full $6s$ subshell – which pulls closer to the nucleus as there is less shielding. This means that when mercury tries to bond, its outer shell (valence) electrons are too attracted to the nucleus to fully delocalise, leaving a liquid.



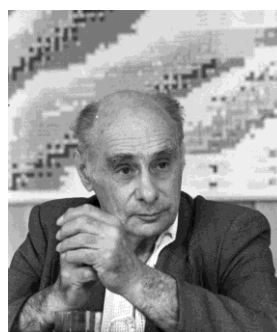


Normal Metallic Bonding

You may – rightly – ask why gold and thallium are not, therefore, also liquids as mercury's neighbours on the periodic table. This answer comes from electronic configuration too! Gold does not have a full outer subshell ($6s$) so can gain an additional electron easily when bonding, whereas thallium has an extra $6p$ electron that is not as strongly attracted as the $6s$ electrons and thus it can experience metallic bonding as well.

To quickly brush over bromine – as it really is not anything special – it follows the melting point trend of its group, with fluorine and chlorine at the top of the group as gases at STP and iodine and astatine below as solids. This trend is a result of decreasing nuclear attraction going down the group – exactly the same reasoning as for mercury.

Now for the exciting part: copernicium and flerovium. Discovered in 1996 and 1999 respectively, there has not yet been enough of these metals synthesised to determine exactly what states they exist in, so we can only make predictions (Linus Pauling would really be in his *element*). Again, when we make predictions, they all hinge around the electronic configurations and our exacting STP – the real unknowns are just how the higher electronic subshells behave. As we have not yet seen these subshells we have to turn to quantum electrodynamics (QED) to make our predictions, however, luckily for the scientists involved QED is the longest-established quantum (field) theory!



Named in Honour; LTR: Nicolaus Copernicus, Georgy Flerov

Once all of the values have been plugged into QED, we find that the melting points of copernicium and flerovium are $10\text{ }^{\circ}\text{C}$ and $-73\text{ }^{\circ}\text{C}$ (respectively) with both boiling points above our STP (Helmenstine, 2020) – so, liquid, right? Yes, in theory, in practice only time will tell whether our theory accurately describes this scenario or if nature will throw us yet another curve ball...

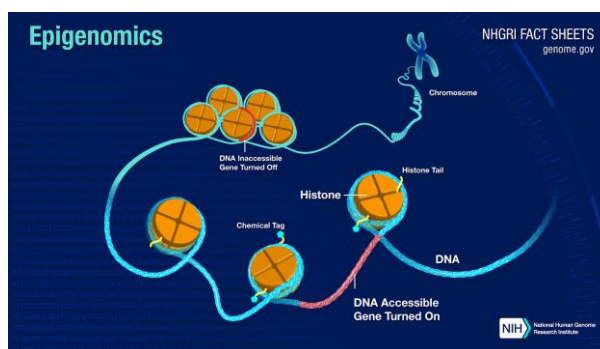


BIOLOGY

MORE THAN JUST YOUR GENES: THE EPIGENOME.

According to the National Institutes of Health in the USA, the genome is the “entire set of DNA instructions found in a cell” (Green and NHGRI, 2024). Your genome encodes for all the proteins in your body (the proteome) and ‘functional’ RNA such as tRNA which taxis amino acids around in protein synthesis. You may know about genes, but what if I told you that two people with the same genome could have different phenotypes - physical expressions - of those genes? This is the *epi*-genome, coming from the Greek word for *above*. The epigenome marks and modifies the DNA, turning genes ‘on’ and ‘off.’ Much of the modification occurs during early development and tissue differentiation, but alterations continue throughout a person’s lifetime. This is important to allow cell specialisation, so cells only synthesise certain proteins.

Now we have a basic idea of what the epigenome is, we can discuss how it works. Firstly, methylation is the addition of methyl (CH_3) groups to bases in the DNA sequence by methyl transferases - ‘-ase’ equals enzyme - either *de novo* to unmodified DNA (e.g. during embryonic development) or during normal DNA replication after development. In particular, for mammals, the methyl groups are added to the 5th carbon of cytosine (recall the four bases ATCG) to form 5-methylcytosine. This reduces gene expression by recruiting gene-repressing proteins or inhibiting transcription factors, which promote RNA polymerase binding to DNA so it can be copied onto mRNA for protein synthesis.



Histones Lock Away Naughty Genes and the Good Ones Get To Come Out and Play!

Second, eukaryotic (anything with a nucleus) DNA is wrapped around histone proteins forming tightly wound spools of DNA that fit more easily into chromosomes in the nucleus; this DNA-histone mix is called *chromatin*. Just like DNA, histones can be modified as part of the epigenome. For example, removing acetyl groups (CH_3CO) from the proteins increases their positive charge, making histones more attracted to DNA's phosphate backbone (PO_4^{3-}). This causes it to ‘wind’ and become more constricted, reducing gene expression because transcription factors and RNA polymerase cannot access the DNA to bind to it.

Importantly, genes are not turned completely ‘on’ or ‘off;’ rather, their ‘volume’ is turned up or down. Also, this is not an exhaustive list of mechanisms - did you think it would be that simple? This is genetics we are talking about! There are other methods, such as folding of chromatin, attachment to the nuclear matrix (the inside wall of the nuclear envelope), methylation of histones themselves and phosphorylation.



Now for the fun part - how it affects us. Surprisingly, 1.5% of the human genome codes for around 21,000 proteins (Lander, 2011), what does the rest of it do? For years, this 'junk' DNA was thought to be just that, but now we are beginning to see otherwise, dubbing it 'non-coding' instead. This comes in many forms: small snippets of non-coding introns between the 'useful' exons in genes or repeating sections of DNA, long and short, throughout the genome. Some codes for functional RNA; some is virus and retrovirus DNA from infections during our evolution that have hitched a ride ever since. These viruses can insert their RNA genome into ours using reverse transcriptase, converting it back into DNA. These acquired genes are sometimes 'jumping genes' or *transposons* linked to changes in DNA position and epigenetics, especially during environmental challenges, but that deserves an article in itself. Bulk methylation silences much of this non-coding DNA, putting it up in the loft for storage, otherwise, we would start synthesising ancient viruses, wreaking havoc on ourselves – it is a good thing so little of our genome is expressed.

Next up, superheroes! Or rather, supermice. The *agouti* gene in mice has *epi*-alleles (different expressions of the same gene) including brown or yellow fur. The yellow phenotype is associated with an increased risk of adult-onset obesity, diabetes, and tumorigenesis (tumour formation). Researchers at Duke University in the USA fed mothers from a long line of yellow agouti mice a diet supplemented with genistein, a major isoflavone in soy, before, during and after gestation. This phytoestrogen induced hypermethylation at 6 sites in the gene, shifting the offspring's phenotype towards the brown coat of wild-type mice and overcoming their obese life sentence. Nurture: 1. Nature: 0. Other work demonstrates epigenetic impacts of parents' diets in human children, especially during early embryonic development. Outside of nutrition, we know epigenetic changes caused by smoking are inheritable even by your grandchildren.



Varying Expressions of the Agouti Gene From 'Viable Yellow' to Wild-Type Brown Fur (LTR)

As well as a diagnostic tool, work has been done in mice on a one-off treatment for acquired and genetic hypercholesterolaemia - or in English, high levels of low-density cholesterol linked to high blood pressure, risk of strokes, and heart attacks. Daily statin tablets, the current treatment, can be forgotten and expensive to manufacture whereas a single-dose treatment removes such issues. With this in mind, researchers targeted the *Pcsk9* gene, involved in cholesterol homeostasis to promote the degradation of low-density lipoprotein receptors on liver cells - hepatocytes. The gene was selected using chemicals known as 'zinc-finger proteins' which interact with DNA as transcription factors do; this repressed *Pcsk9* in hepatocytes by 50% over a 1-year period when the trial ended (Dolinoy et al., 2006). Therefore the treatment was considered effective into mouse adulthood and there is promise for similar 'one-off' treatments.



PHILOSOPHY

THE COLLAPSE OF THE WAVEFUNCTION

Quantum theory forms the cornerstone of modern physics, but it has a serious reality problem. But then, that is probably to be expected from a field where particles can have neither a certain position nor a certain momentum.

Central to quantum theory – in this case quantum field theory (QFT) – is the concept of a wavefunction (ψ), simply put, it is a mathematical function that describes every point in a field at a specific time and, therefore, how these points change with time. The way it describes each point is with a probability: the likelihood of finding a particle at that point. So, by making calculations of the wavefunction, we can find the probabilities of particles moving in specific ways over time.

$$i\hbar \frac{\partial}{\partial x} \psi = H\psi$$

Schrödinger's Wave Equation

Never fear! We will not be doing any mathematics today – as it happens, the maths is rather complicated, but that is beside the point – we will merely be looking at the philosophical implications presented by the mathematical framework itself. You see, the framework deals entirely with probabilities so, whilst there may be a high possibility of a certain outcome, we cannot be certain until we make a measurement of the quantum system. We call this measurement event the ‘collapse of the wavefunction’ as it is at the point of measurement that all of the probabilities appear to ‘collapse’ into a real outcome.



QFT Considers Probability Waves Like Water Waves

You have to understand at this point that the probabilities we use are not because of a lack of knowledge. In a coin flip, if you know *everything* about the event, you can determine the outcome with certainty – this is not the case with a quantum system, the probabilities are an inherent part of the system, no amount of knowledge can ever allow us to know the outcome. To reinforce this point, let me provide an example of how we could discover this fact; set up a system whereby we know everything about a particle and observe it one second later. Now, if we repeat this measurement on identical systems we will observe that the particle ends with a different final configuration – the probabilities are truly a part of nature.



To make this worse, many quantum physicists consider all of these probabilities as *actually* happening until a measurement is made – in what is known as a superposition of states, quantum objects (quons) take on all possibilities at the same time. Imagine, again, flipping a coin, if you do not look at the coin once it lands you have no way of knowing whether it landed on heads or tails (numismatists, that is on the obverse or reverse, I do apologise). In this case, it would be a fair assumption to consider that either option has happened – we simply do not know – and this is where superposition emerges, we cannot make any assumptions, so we say that every possibility has happened until we make a measurement.

This measurement problem is where the solid physics gives way to the slippery slope of philosophy. Try now to think about at which point the wavefunction ‘collapsed’ and the messy probabilities gave way to our neat reality of one real outcome. With nature only providing us with limited information – that is where our probabilities come from – we cannot say that the outcome was predetermined, like Newton’s grand ‘clockwork universe.’ But, neither can we say that the outcome is only determined once we make a measurement, that is a simply preposterous notion. And either way, when does the measurement actually take place, is it in our instruments? Or is it in our eyes when we observe the result?

This is starting to seem as though the problem is raising more questions than it is answering and, indeed, that is the nature of philosophy – there is no *correct* answer. Niels Bohr and Werner Heisenberg produced, in my mind, the perfect interpretation for a physicist: why should we care? Known as the Copenhagen (København) interpretation, Bohr and Heisenberg simply state that “there is no deep reality” (Herbert, 1985), in other words, the dynamic attributes (probabilities) we give to quons are not a part of nature and that quons do not take on any definite attributes until we measure them. Essentially, our elaborate QFT framework describes the measurement event, not what comes before.



Dapper Chaps; LTR: Niels Bohr, Werner Heisenberg

Physics is developed to fit the facts we find, as with all of science, and we simply cannot assume that the frameworks and models we develop describe those things outside of observation. Sure, it would seem mighty strange if the book on my desk travelled half the way around the world and back when I am not looking at it – but it does not mean that it does not happen. Sadly, there is simply no way of knowing what quons get up to whilst we are not looking. But, it does not hurt to postulate the true feeling of the universe.



NOMENCLATURE & METROLOGY

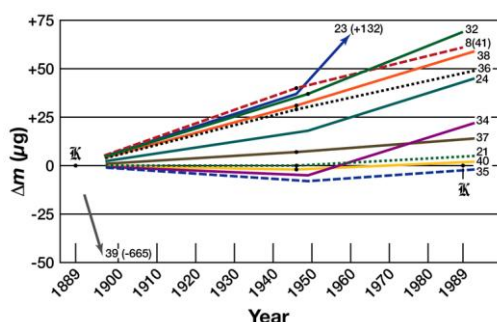
THE KILOGRAM IS DEAD, LONG LIVE THE KILOGRAM

“I’m going to make you an offer you cannot refuse,” Big K said to the world in 1889. Far from being a Mafia boss, Big K (or Le Grande K, **K**) was to be the world’s mass standard for over 140 years. Forged in 1879 out of a platinum-iridium alloy and stored in an underground vault on the outskirts of Paris, the International Prototype of the Kilogram (IPK, **K**) was the only standard of *exactly* one kilogram. Imagine our surprise then, when we discovered that it was no longer quite a kilogram – crisis!



Big K (The IPK)

At the same time as **K** was manufactured, 40 other masses were produced from the same alloy in the same way – making them identical. These masses were distributed to countries across the world for use in national laboratories and would set the standard for everything from drug production to bathroom scales. However, the 1889 resolution that established **K** as the mass standard also required that these 40 masses be assembled and measured every 40 years to determine whether the standard was still accurate.



The Standard Was Not Accurate

Over time, it became apparent that the 40 masses were slowly drifting away from **K** in as much as 68 μg (micrograms) and roughly 50 μg on average – about the weight of a human eyelash (Resnick, 2018). This may not sound like a lot, but when some drug measurements are made in micrograms, it can be a life-and-death situation! You may, rightly, say: ‘Well, **K** stays the same so just use that,’ this is a big mistake. **K** sets the standard and so will not be observed to change from the standard – it is a problem of self-reference, and we now have no idea which of the 41 masses is closest to the original 1889 mass. There are many hypotheses for why this



mass drift has occurred, everything from corrosive cleaning solutions to dust in the air may well have affected the masses. The solution? Do not rely on a physical object!

On the 20th of May 2019 (World Metrology Day – awesome), a new standard for the kilogram was chosen:

The kilogram, symbol kg, is the SI unit of mass. It is defined by taking the fixed numerical value of the Planck constant, h , to be $6.626\,070\,15 \times 10^{-34}$ when expressed in the unit J s, which is equal to $\text{kg m}^2 \text{s}^{-1}$, where the metre and the second are defined in terms of c and $\Delta\nu_{\text{Cs}}$.

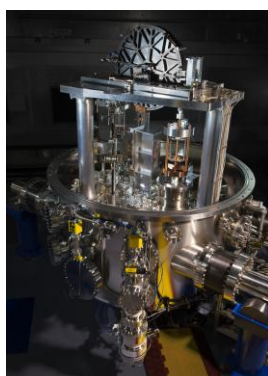
(BIPM, Bureau International des Poids et Mesures, 2019)

Simple. Only joking, standards are hardly ever simple, so where does this come from? The new kilogram standard relies upon the standards for the second and the metre, of which the second is defined using a caesium atom and the metre is defined using the second and the speed of light. This should emphasise how every unit is now built on top of one another using a system of fundamental constants – in fact, there are seven constants across seven units (Newton, much). The kilogram introduces the Planck constant as a way of connecting the kilogram to the second and metre as follows:

$$1 \text{ kg} = \left(\frac{h}{6.626\,070\,15 \times 10^{-34}} \right) \text{m}^{-2} \text{s}$$

Presenting: the Kilogram

What this equation essentially says is that scientists have fixed the value of the Planck constant and, since we know what the second and metre are, the only undefined part of the equation is the kilogram which remains after dividing the Planck constant by itself. At this point, those of you with sharp minds will have spotted a problem.



The Kibble Balance

How did we fix the value of the Planck constant? The National Institute of Standards and Technology (NIST) in the USA created a piece of equipment called a Kibble balance which took the basis of an old-style mass balance and replaced one of the pans with a solenoid in a magnetic field. This coil can exert a force on the system and balance the calibrated mass of Big K and, since the applied force can be written in terms of the Planck constant, allowed scientists to fix the exact value of the Planck constant to make our kilogram definition work.

ARTICLE BY NSR, INSPIRED BY AHT



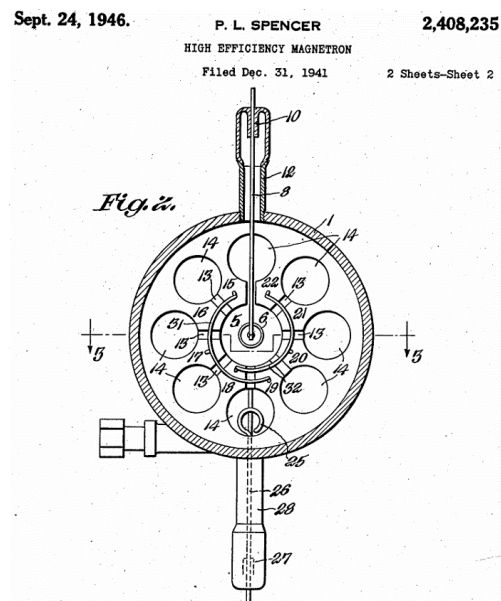
ENGINEERING

SMILE AND WAVE

In the early 1940s, a radar engineer was testing a magnetron when he reached into his pocket and felt a strange warm fluid. This was his chocolate bar, a snack he was rather looking forward to but was now a gooey mess – why? He was a victim of the first microwaves.

The engineer in question was Percy Spencer, working at the Raytheon Company on magnetrons that produced microwaves for radar systems responsible for detecting German U-boats and bombers in the British Second World War effort. Spencer trialled these magnetrons on a variety of food from eggs to popcorn, making careful observations of cooking times, which turned out to be much faster than those of a traditional oven.

But, what is a magnetron? For those that remember, a useful image to conjure is of a cathode ray tube similar to that in the first form of television set common before the millennium (2000). In a cathode ray tube, a metal plate (cathode) is heated to produce electrons via thermionic emission, these electrons are then accelerated by a charge difference from an anode ring positioned in line with the cathode. The electrons are accelerated towards a phosphor screen and bent into place using a strong electromagnet where the phosphor gains energy from the electron and glows to produce an image.



Spencer's Microwave Oven Magnetron Design Patent (Fig. 2.)

In a magnetron, the cathode ray tube design is modified to put the cathode as a central rod (Fig. 2. 6) with the anode surrounding the length of the cathode with a small air gap (5). In the anode there are a number of slits (13) that lead to cavities (14) – these will be important later. The other big change is that the whole assembly is sandwiched above (into the page) and below (out of the page) by the poles of a permanent magnet.



This setup forces electrons to accelerate in a circle around the air gap (5) and as electrons pass the slits (13), the cavities (14) resonate to produce microwave radiation which is funnelled into a microwave oven or the air through a radar dish using a waveguide.

As I may have hinted, a magnetron is the central part of a microwave oven, the technology pioneered by Percy Spencer, and the diagram I used came from his very patent filed on the 31st of December 1941 – this design would go on to change the world. To complete the assembly of the electronics, we need a high-energy power source which comes in the form of a transformer and provides the current to heat the cathode and create the large potential (charge difference) between the cathode and anode.



The Safety Cage; LTR: A Faraday Cage in Action, a Really Fancy Faraday Cage

We now have microwaves, and we will store these in a Faraday cage (big metal box) to make sure they do not burn us! But, there is one more thing, a rotating glass plate – ah, the ubiquity. The plate makes sure that food is cooked evenly by rotating it through the microwaves, this is because the microwaves form standing waves inside a Faraday cage and thus we have cold spots and warm spots aligning with the peaks and troughs of the standing waves, the difference between peaks and troughs is a significant 6 cm, half the length of a microwave used in a microwave oven (Woodford, 2021). Without the rotation, your frozen lasagne would remain freezing cold in some spots and be boiling hot in others!

Only one question remains, just how exactly do waves cook food? The microwaves in the oven pass straight through your food and deposit some of their energy in the molecules of your dish of choice. This energy comes in the form of kinetic energy which causes the molecules to vibrate more rapidly and the faster the vibration, the hotter your food. This does not, as many people will say, cook food from the inside out, rather it heats all of the molecules in your food at the same time.

This equal heating ensures that food is cooked evenly and allows for the faster cooking times that Spencer initially found. An interesting fact is that the water molecules in food experience more of the heating effect than other molecules so food like apple pie with lots of water molecules in the middle may very well cook from the inside out – oops, myth not quite dispelled!



MEDICINE

RASE-ING A MUTANT: PCR IN CANCER DIAGNOSIS

In 1983, Dr Kary Mullis revolutionised molecular research with the Polymerase Chain Reaction (PCR) protocol. PCR was the ‘gold standard’ test of the COVID-19 pandemic response and has sowed seeds as a diagnostic tool for cancer since the 90s. Let us take a step back before diving into that and look at cancer; the uncontrolled division of cells which can invade neighbouring tissues or spread through the blood or lymph system. Usually caused by direct mutations or changes in the expression [see Biology article] of genes that otherwise prevent cancer, 1 in 2 of us are destined to develop it (NHS, 2022). We all hear about the common treatments: chemo-, radio- and newer immunotherapies - but how do we diagnose cancer in the first place?

Oncologists have access to all the toys modern medicine offers. Complete blood counts and chemistry indicate early tumour development, and imaging such as a PET scan locates tumours by tracking glucose accumulation. Biopsies (tissue samples), and investigative surgeries would directly look for tumours. Cytogenetic analysis (changes in chromosomes), and immunohistochemistry can identify cancer, and plan and monitor treatments. This is by no means an exclusive list and you might think this is all very complicated - because it is. One thing is crystal clear: early diagnosis massively improves prognosis (prediction). Therefore, let us focus on immunohistochemistry or ‘IHC’.

IHC is essentially a broader, clinical version of the ELISA test in AQA A-Level Biology. Antigens (chemicals protruding on the surface of cells) are first ‘retrieved’ from a tissue sample by proteolytic (protein-digesting) enzymes and a buffer or heat, the antigens would otherwise be undetectable as the samples are fixed in formalin and embedded in paraffin wax. Next, antibodies are applied and bind to the antigen as they are *complementary* to the antigen, meaning they fit almost like a lock and key.



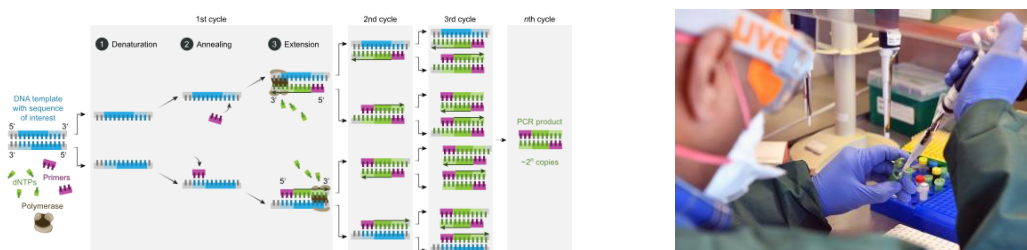
LTR: Paraffin Cases Solidify Post-Production Before Tissue Preparation, Antibodies Floating Around Not Doing Very Much

As an aside, the antibodies used are mono- or polyclonal. Monoclonal antibodies originate from many clones of *one* cell. Polyclonal antibodies come from an entire organism after inoculation (like with a vaccine), obtained directly from the organism’s purified serum (blood plasma) - therefore originating from many cells. This is important clinically as polyclonal antibodies have greater versatility and cost-efficiency than monoclonal antibodies which offer incredible precision and specificity.

Back to IHC, generally, another set of antibodies is applied to bind with the primary antibodies. To provide a semi-quantitative result, these secondary antibodies are labelled either with



a fluorescent molecule or an enzyme such as horseradish peroxidase. Enzyme labelling produces a coloured product after incubation with a chromogenic (colour-producing) substrate. This increases sensitivity because a few enzymes can cause large colour changes. However, like all things, IHC has its pitfalls. It suffers from error risks such as interfering enzymes from the tissue sample, antigen over or under-fixation, air bubbles, and residual haematopoietic elements (blood contents). Enter, PCR...



LTR: PCR Doing Its Thing, Putting Chemicals in a Tube and Calling It Science

Malignant cells shed DNA fragments and other molecules into the bloodstream, dubbed tumour markers. PCR takes these DNA snippets in a sample and replicates them to a quantity detectable by methods such as electrophoresis - look it up! The DNA template derived from the sample is added to a buffer with nucleoside triphosphates (nucleotide precursors). Also added are, DNA-like molecules called oligonucleotide primers that bind either side of the gene(s) being searched for. These mark the gene(s) for replication by DNA polymerase in the buffer, acting like starting blocks in the 100 m nucleotide sprint.

The mixture is heated to denature (separate) the sample DNA strands and cooled to allow the primers to anneal (join) with the template. The polymerase extends the oligonucleotide primers to produce a new, complementary DNA strand, doubling the number of strands of the target DNA sequence. With each thermal cycle, the primers in the buffer anneal to the new and original DNA strands using both as templates.

In this way, the target gene(s) are exponentially amplified to detect the presence of an incredibly small quantity of tumour markers as well as physical alterations of genes, oncogenic (cancer-causing) viruses or expression of cancerous genes. A group of researchers found that 'A single cell expressing a tumour marker among 10-100 million lymphocytes [white blood cells] can be detected by the PCR assay' (Raj, Moreno and Gomella, 2000), indicating the organ of origin, stage and type of cancer, often before symptoms arise.

This also can be used *after* diagnosis to test for genetic alteration and inform treatment, often working in tandem with IHC. One example is non-small cell lung cancer (NSCLC). IHC is used to test for PDL1 expression; this gene suppresses T-cell activity in the immune system. T cells hunt cancerous cells so cancer caused by its expression means Pembrolizumab immunotherapy will release the brakes, rase-ing the mutant cells. Next Generation Sequencing, the clinical term, uses polyme-*r*ase chain reaction (get the pun yet?) to test for an array of aberrant genes, again allowing treatment to target these genes accordingly; modern black-ops versus traditional carpet bombing of chemo- or radiotherapy.



EDITORS MESSAGE

2023-2024

Although this newsletter has not yet been published for a year, it is the end of an academic year and, in my mind, this is the ideal time to both look back and discuss the future.

This Year

I have been overwhelmed with the support I have received for this newsletter since I first published in April this year – the response has been greater than I ever expected! But, the plan for this newsletter was never for it to be a *singular* effort, and this year I have been joined by an excellent biology writer who has managed to fit into this newsletter perfectly – I thank them for their contributions and look forward to what they write next.

Moreover, I would also like to thank those of you who have written or spoken to me about this newsletter, every comment has been heard and I hope that you can see the effects of your kind words – especially those of my content advisor (yes, I did give you a title!) who has provided endless suggestions for articles and is the reason for the engineering article in this very issue: *thank you!*

Next Year

Whilst on my travels this year I have experienced an extreme heat wave in Italy, a dry rainy season in Sri Lanka and a month of heat in Albania where there should have been just two weeks. We hear stories like these from across the globe: coastal towns sinking into the sea, animals suffering from unexpected droughts and more frequent tropical storms destroying cities. This is the result of our rapidly changing climate and whether its evidence comes from Greta Thunberg, Donald Trump, or your mate Dave at the pub – the good, the bad *and* the ugly – we must listen to and heed the warnings, we are heading toward catastrophe.

To raise awareness for the climate crisis and encourage you to look for solutions – everything from lobbying governments to changing habits – I will be including at least one climate-based article in every issue next year (highlighted with this symbol: 🌱) which will appear as one of the main articles. These articles will present some of the issues or solutions to climate change, and, however outlandish they may seem, they are there to make a point – we should not sit and do *nothing*.

Moving on, next year, I will still welcome all suggestions and corrections you may have – none is too small or too large! And if you would like to write an article for this newsletter, I would be more than happy to include it in a future issue. I also welcome any and all research or products from the recommendations – I always enjoy people engaging with these newsletters!

So, here's to next year 🍷



NSR (THE EDITOR)



OCN (WRITER)



RECOMMENDATIONS

A-LEVEL WIDER INDEPENDENT STUDY

- 📖 **Mathematics:** The Golden Ratio and Fibonacci Numbers.
- 📖 **Physics:** What Is Absolute Zero?
- 📖 **Chemistry:** The Lab on a Chip.
- 📖 **Biology:** Transcription Factors and Metamorphosis.

DISCRETE FURTHER READING

- 📖 **Mathematics:** *The Mathematical Intelligencer June 2024*, A Bayesian Treatment of the German Tank Problem ➡
- 📖 **Physics:** *Physics World August 2024*, Smashing heavier ions creates superheavy livermorium ➡
- 📖 **Chemistry:** *ChemistryViews July 2024*, Tales of Olympic Marathons ➡
- 📖 **Biology:** *Nature January 2024*, Forget lung, breast or prostate cancer: why tumour naming needs to change ➡

CONTINUOUS FURTHER STUDY TOPICS

- 📖 **Mathematics:** Is the Painter's Paradox Really a Paradox?
- 📖 **Physics:** Spectroscopy
- 📖 **Chemistry:** The Lanthanide Contraction
- 📖 **Biology:** Gene Drive Engineering

BOOK APPRAISAL

Parker, M. (2024). *Love Triangle*. [online] Allen Lane. Available at: amazon.co.uk/dp/B0CFXCP3YN ➡.

SUMMER PROBLEM

Solve for x :

$$2\log_2 x - \frac{1}{2}\log_2 x = 3$$

Solution next month.

Last month's solution:

$$x = -1$$



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See *Editors Message 2023-2024*.

CORRECTIONS

- ✉ **Issue 3 - Physics:** The first edition distributed contained a rendering artefact which prevented some parentheses from being displayed. This has been corrected in the final digital edition.

The editor can be contacted regarding errors and suggestions via the email skinners-science-spectrum@magna-aspirations.org .

Join us next month in celebrating CERN's 70th anniversary, I really want a T-shirt...



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